

Seeking Demand Curve Experiment Report

1. Experimental background

This experiment simulates a perfectly competitive market in which the traded good is a backpack. In the experiment, the teacher plays the role of the backpack supplier, while students participate in the auction as market demanders and make bids according to their true willingness to purchase the backpack.

Through this experiment, we can observe how the quantity demanded by students changes at different price levels and use the experimental bidding data to construct the demand curve for the backpack market. At the same time, the experiment also requires us to calculate the transaction price that maximizes the supplier's revenue based on the principle of revenue maximization.

In this experiment, the transaction cost is assumed to be zero. Therefore, the supplier's revenue can be expressed as follows:

$$\begin{aligned} \text{Supplier's revenue} \\ &= \text{Transaction price} \times \text{Transaction quantity at that price} \end{aligned}$$

The core objective of this experiment is to analyze the demand curve based on the bidding data and identify the optimal price that maximizes the supplier's revenue.

2. Experimental content and steps

This experiment was conducted using an auction method. At the beginning of the experiment, the teacher first displayed the auction item, namely a specific backpack, and explained that the price of the backpack was only a nominal price used in the experiment. Students did not need to bear any real transaction cost.

Then, the teacher explained the experimental rules to the students. Students were required to make bids based on their true preference for the backpack and their willingness to pay. The bids should be reasonable, normally not exceeding 500 yuan and not lower than 0 yuan. Meanwhile, students were not allowed to collude with each other; otherwise, the related bids would be regarded as invalid.

During the bidding stage, students filled in their own auction bids. After all bids were completed, the teacher collected and organized the bidding data. During data processing, unreasonable extreme bids, such as bids of 0 yuan or below and bids of 500 yuan or above, needed to be removed.

After that, all valid bids were sorted in descending order of price. For each bid price, the number of students who were willing to buy the backpack at that price or above was counted. This number represents the possible transaction quantity at that price. Then, the supplier's revenue at each price level was calculated using the following formula:

$$\text{Total revenue} = \text{Price} \times \text{Transaction quantity}$$

Finally, the demand curve was drawn based on the organized bidding data. By comparing the total revenue at different price levels, the price that maximized the supplier's revenue was identified.

3. Experimental results (Demand Curve) and your bid price

In this experiment, my bid price was 180 yuan.

Based on the bidding data collected in this experiment, all valid bids were arranged in descending order. Then, the cumulative transaction quantity and the supplier's total revenue at each price level were calculated.

The demand curve takes transaction quantity as the horizontal axis and bid price as the vertical axis. According to the experimental data, as the price gradually decreases, the number of students willing to purchase the backpack gradually increases. Therefore, the demand curve obtained in this experiment generally shows a downward-sloping trend from left to right. This is consistent with the general law of demand: the higher the price, the smaller the quantity demanded; the lower the price, the larger the quantity demanded.

Some key revenue results from the experimental data are shown below:

Price / yuan	Transaction Quantity	Supplier' s Total Revenue / yuan
175	20	3500
168	22	3696
157	24	3768
150	25	3750
150	26	3900
150	27	4050
150	28	4200
136. 42	29	3956. 18

The data show that when the price is 150 yuan, there are 28 students willing to buy the backpack at this price or above. At this point, the supplier's total revenue is:

$$150 \times 28 = 4200 \text{ yuan}$$

This is the maximum revenue among all price levels in this experiment. Therefore, the optimal transaction price corresponding to the supplier's revenue maximization is:

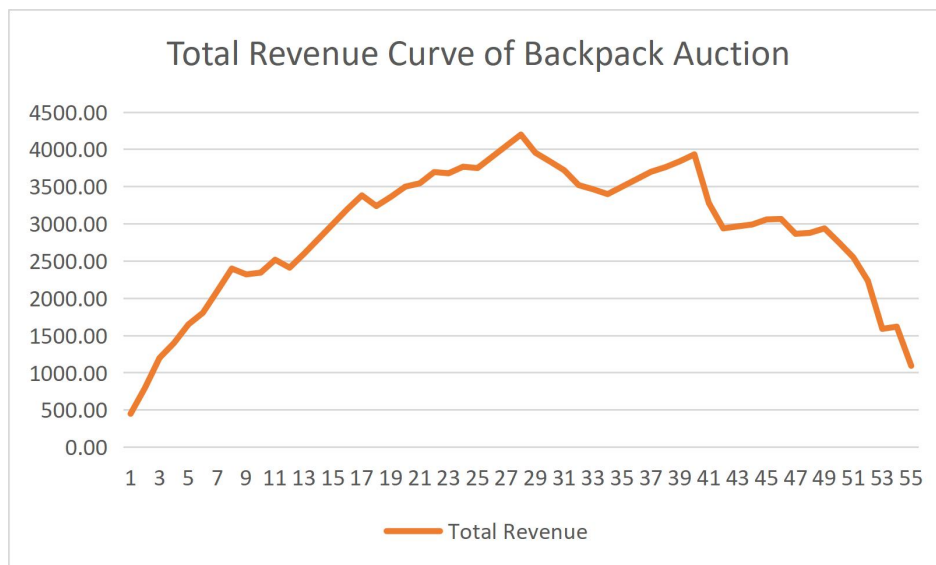
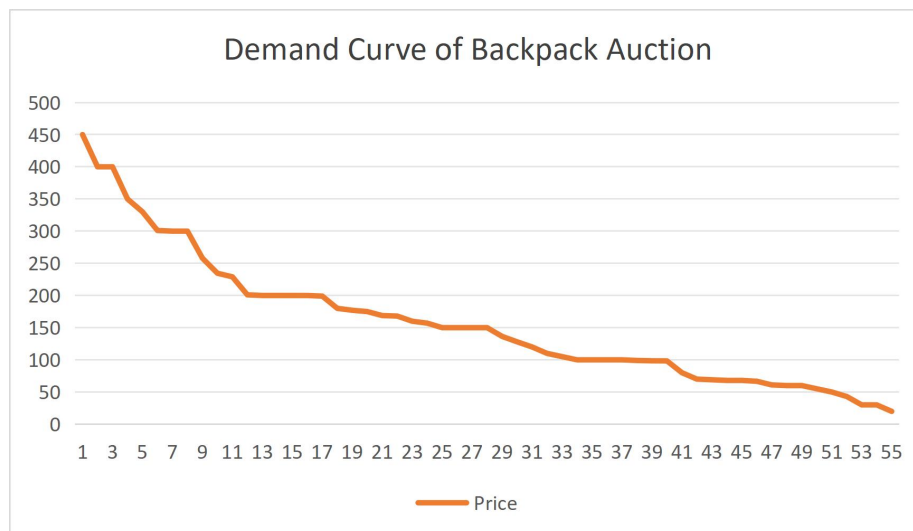
$$\text{Optimal transaction price} = 150 \text{ yuan}$$

$$\text{Maximum revenue} = 4200 \text{ yuan}$$

$$\text{Transaction quantity} = 28$$

Therefore, the conclusion of this experiment is that when the backpack price is set at 150 yuan, the supplier can obtain the maximum potential revenue of 4200 yuan.

The following two figures show the demand curve and the total revenue curve based on the experimental bidding data. The demand curve shows the relationship between price and transaction quantity, while the total revenue curve is used to identify the revenue-maximizing price.



4. Learning Experiences in this Experiment

Through this experiment, I gained a more intuitive understanding of how a demand curve is formed. A demand curve is not obtained only from theoretical assumptions; it can also be constructed from actual consumer bidding data. In this backpack auction experiment, each student's bid represented his or her willingness to pay for the backpack. After sorting these bids from high to low, we could obtain the quantity demanded at different price levels.

This experiment also helped me understand the relationship among price, quantity demanded, and supplier revenue. When the price is high, although the selling price of each backpack is higher, fewer students are willing to buy it, so the total revenue may not be high. When the price is low, although more students are willing to buy it, the unit selling price decreases, which may also reduce the total revenue. Therefore, the supplier needs to find a balance between price and transaction quantity instead of simply choosing the highest or lowest price.

According to the experimental data, when the price is 150 yuan and the transaction quantity is 28, the supplier's total revenue reaches the maximum value of 4200 yuan. This indicates that 150 yuan is a relatively suitable transaction price under the conditions of this experiment.

Through this process, I further understood the relationship among the demand curve, auction mechanism, and revenue maximization, and I also learned how to use experimental data to calculate the optimal transaction price in a market.